MIS 381N: Information Management (05270 and 05275)
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Homework 4 – Document AI

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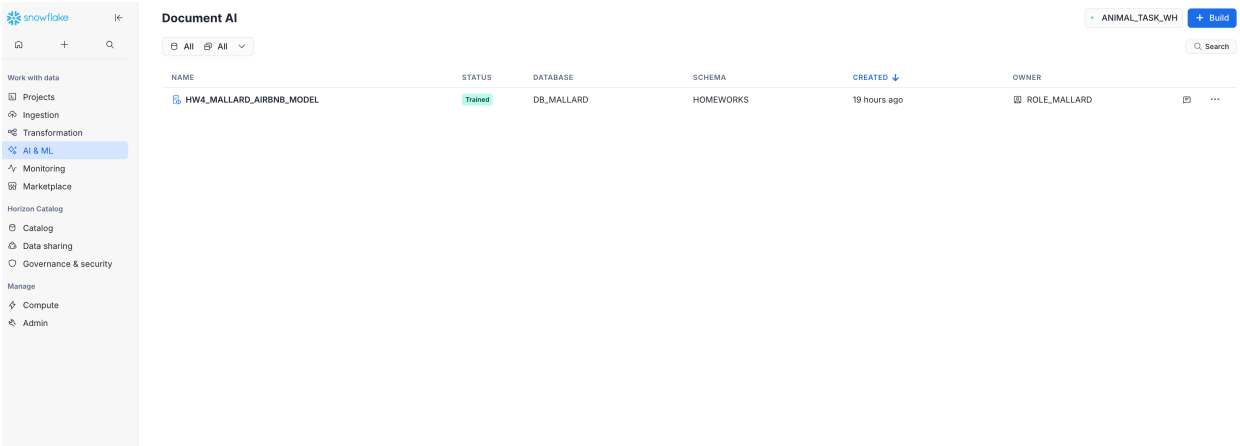
The purpose of this project was to build, train, and evaluate a custom Document AI model in Snowflake capable of extracting structured information from semi - structured Airbnb listing PDFs. The Airbnb property listing of California, for 6 guests from 22nd December to 27th December is chosen for the Document AI project. Using a dataset of 20 Airbnb training documents and 5 unseen test documents, the goal was to design a consistent set of extraction fields, engineer effective prompts, refine model outputs, and validate accuracy across multiple layouts. This project demonstrates the process of preparing datasets, defining entities, tuning prompts, training and publishing a Document AI model, and evaluating performance using real, diverse Airbnb listing formats.

The selected Airbnb documents included variations in layout, formatting, and content placement, making them ideal for testing the robustness of a custom extraction pipeline. By configuring 15 different fields - such as listing title, location, property attributes, host information, ratings, pricing, and cancellation policy - the project showcases the practical application of Document AI for real - world, semi - structured data processing. The workflow also emphasizes the importance of prompt clarity, field definitions, and the ability to handle inconsistent document structures.

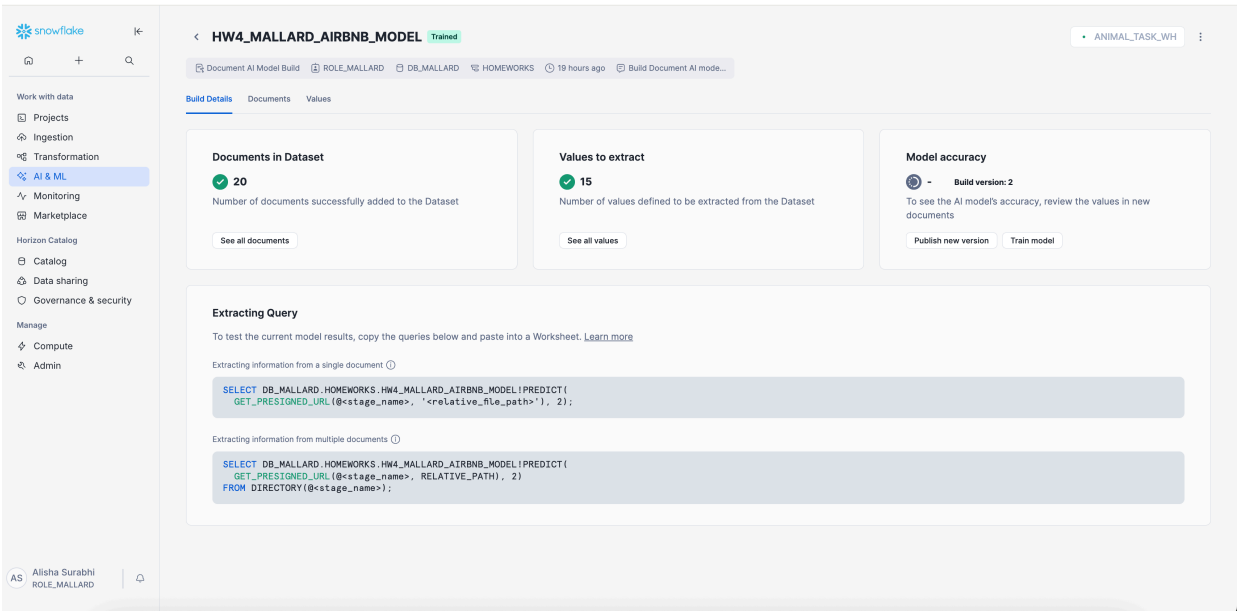
Document AI overview page

The Document AI overview page shows the registered model HW4_MALLARD_AIRBNB_MODEL under the HOMEWORKS schema, with its status marked as Trained. This verifies that the model was created, configured, and saved correctly in the Snowflake environment.



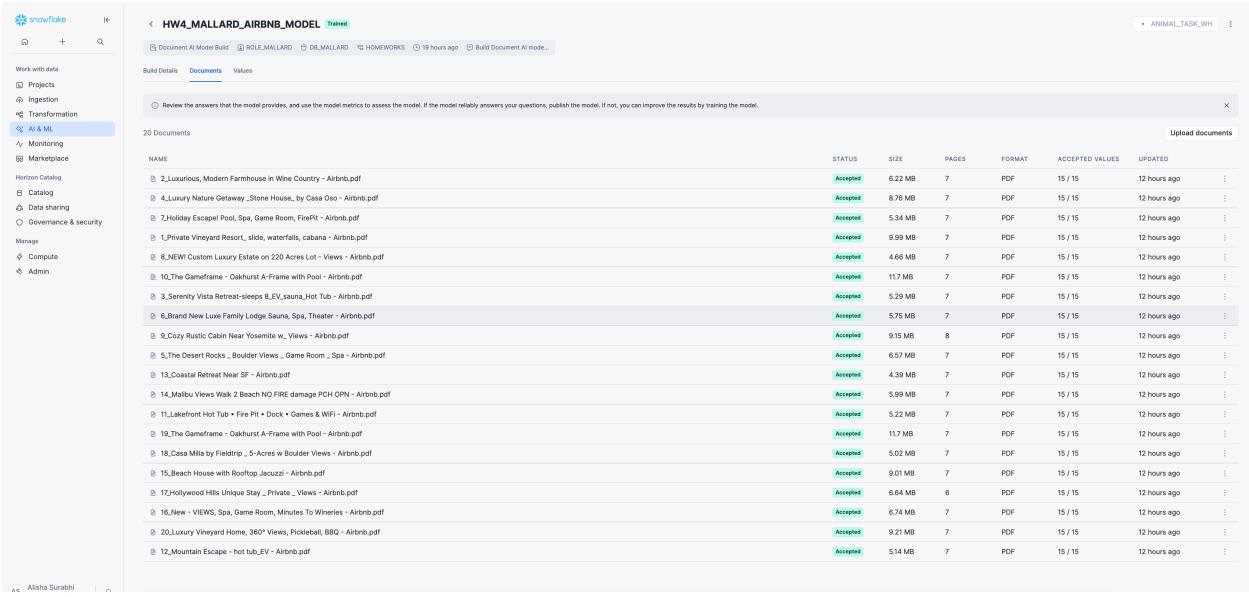
Document AI Model summary page

The Document AI Model summary page verifies successful ingestion of 20 Airbnb PDFs and configuration of 15 custom extraction fields.



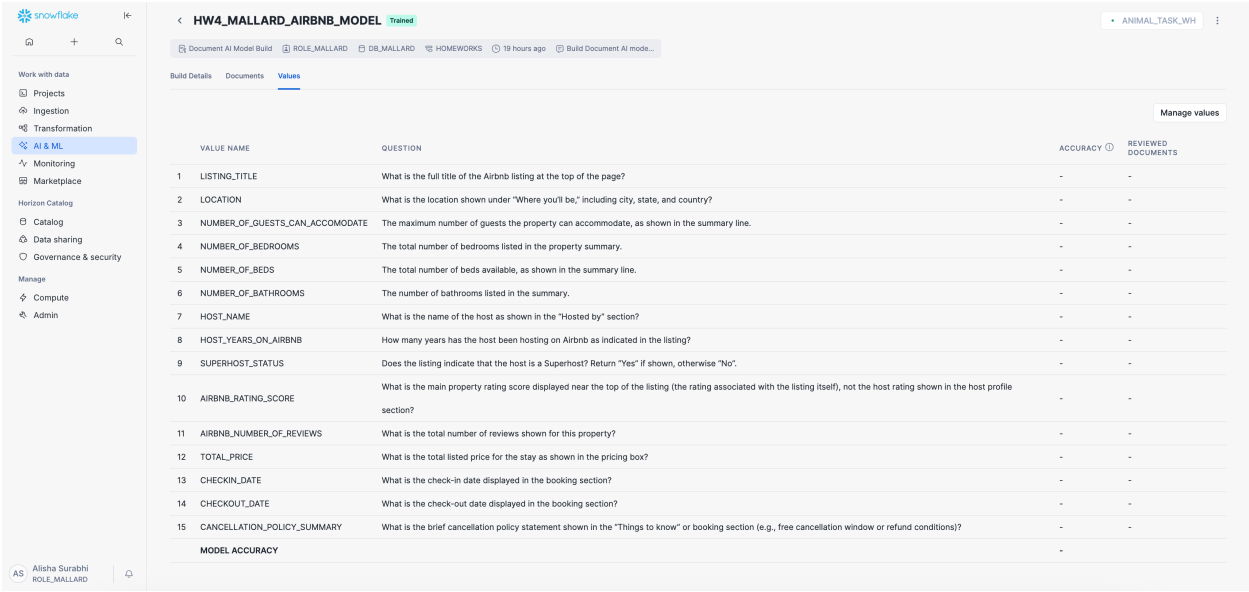
Documents tab

The Documents tab lists all 20 training documents used to build the model, with each marked as *Accepted* and showing consistent extraction of the defined 15 fields. This validates the completeness and reliability of the training dataset for the Document AI model.



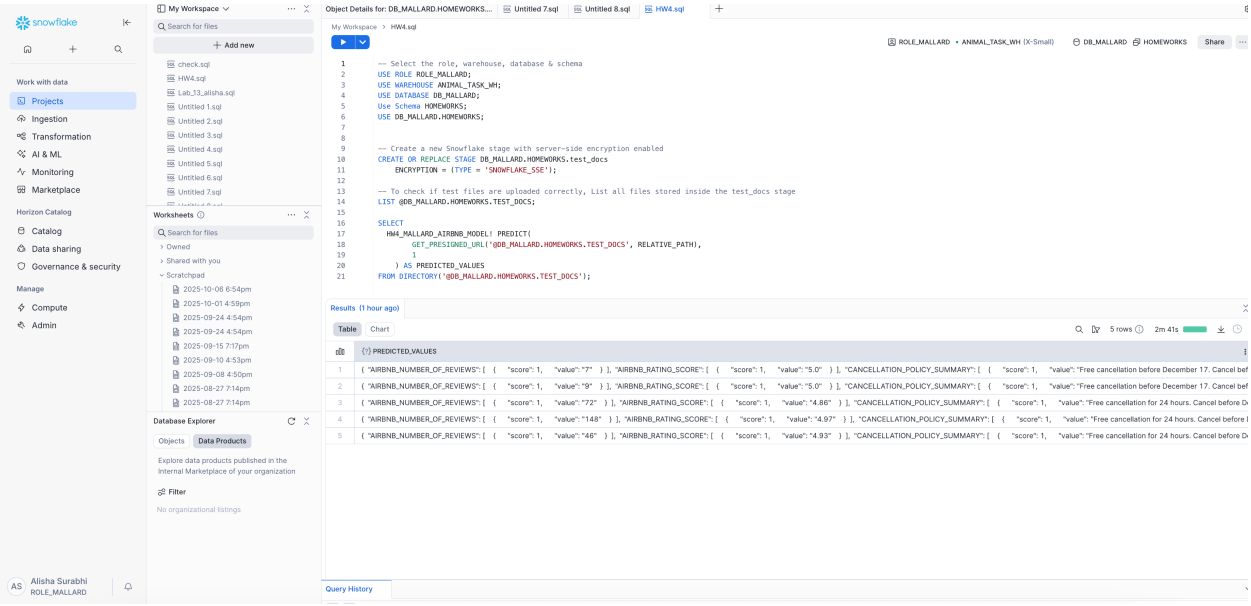
Values tab

The below Values tab shows the full schema of extraction fields, such as title, location, ratings, pricing, and host details, used to train the Document AI model. This view validates that each field was clearly defined with precise prompting to ensure accurate data extraction.



SQL workspace

The below SQL workspace demonstrates the end - to - end prediction workflow, including context setup, stage listing, and model inference. The output section displays the JSON predictions generated by the HW4_MALLARD_AIRBNB_MODEL, validating its deployment and accuracy on unseen documents.



Predicted vs Actual Comparison

Across the five test documents, the Document AI model performed very well, achieving near - perfect accuracy with only one incorrect prediction. Most structural property - level field, NUMBER_OF_GUESTS_CAN_ACCOMODATE, NUMBER_OF_BEDROOMS, NUMBER_OF_BATHROOMS, HOST_NAME, HOST_YEARS_ON_AIRBNB, SUPERHOST_STATUS**,** and even NUMBER_OF_BEDS for four out of five test files, were extracted perfectly. The only mismatch occurred in Test Doc 2, where the model predicted 1 bed instead of the correct value 4 beds. This error appears to stem from layout inconsistency: the page does not explicitly list all bed counts upfront, causing the model to rely on partial text from the bedroom descriptions.

A few fields required prompt tuning during the development process. Both AIRBNB_RATING_SCORE and AIRBNB_NUMBER_OF_REVIEWS initially suffered from confusion between multiple rating elements on the page; in some cases, the model attempted to use host rating instead of property rating, particularly when host profile cards appeared near the top of the layout. After refining the entity descriptions and explicitly instructing the model to extract the property rating only, these fields stabilized and produced perfect predictions across all five test documents.

Two fields - LISTING_TITLE and CANCELLATION_POLICY_SUMMARY - showed minor formatting differences between the PDF text and the model's output. These were not true extraction errors but rather differences caused by layout variations, line breaks, and punctuation changes across the Airbnb template. Despite these differences, all extractions semantically matched the source content.

Price extraction worked exactly as intended. During early testing, the model tended to pick up the strikethrough "original price," so the training dataset was manually corrected to emphasize only the non - strikethrough, discounted price. After training, the model consistently extracted the correct displayed price in all test documents.

Overall, the model achieved 100% accuracy for four of the five test documents and 93% accuracy for Test Doc 2, with the only error being the NUMBER_OF_BEDS field. These results demonstrate that the model generalizes well across semi - structured Airbnb layouts, with remaining inconsistencies largely due to PDF formatting differences rather than extraction logic.

<u>Test Doc 1</u>

Field	Test Doc 1		
	Actual (from PDF)	Predicted (from model)	Match
NUMBER_OF_+E:PBEDS	Serene Views | Hot Tub | BBQ | Fire Pit | Game Room	Serene Views; Hot Tub; BBQ; Fire Pit; Game Room	Yes
LOCATION	Oakhurst, California, United States	Oakhurst, California, United States	Yes
NUMBER_OF_GUESTS_CAN_ACCOMODATE	10	10	Yes
NUMBER_OF_BEDROOMS	4	4	Yes
NUMBER_OF_BEDS	4	4	Yes
NUMBER_OF_BATHROOMS	3	3	Yes
HOST_NAME	Kyle	Kyle	Yes
HOST_YEARS_ON_AIRBNB	11	11	Yes
SUPERHOST_STATUS	Yes	yes	Yes
AIRBNB_RATING_SCORE	5	5	Yes
AIRBNB_NUMBER_OF_REVIEWS	7	7	Yes
TOTAL_PRICE	$3,013	$3,013	Yes
CHECKIN_DATE	12/22/25	12/22/25	Yes
CHECKOUT_DATE	12/27/25	12/27/25	Yes
CANCELLATION_POLICY_SUMMARY	Free cancellation before December 17. Cancel before check-in on December 22 for a partial refund.	Free cancellation before December 17. Cancel before check-in on December 22 for a partial refund.	Yes
Accuracy			100%

The comparison table for Test Doc 1 validates the model's precision, with every field - including listing details, host attributes, and cancellation policy - matching exactly between the PDF and the model output, achieving full accuracy.

Test Doc 2

Field	Test Doc 2		
	Test Doc 2+E:P	Predicted (from model)	Match
NUMBER_OF_+E:PBEDS	Designer Lakehouse-Private Dock!	Designer Lakehouse-Private Dock!	Yes
LOCATION	Bradley, California, United States	Bradley, California, United States	Yes
NUMBER_OF_GUESTS_CAN_ACCOMODATE	10	10	Yes
NUMBER_OF_BEDROOMS	4	4	Yes
NUMBER_OF_BEDS	4	1	No
NUMBER_OF_BATHROOMS	3	3	Yes
HOST_NAME	Haley	Haley	Yes
HOST_YEARS_ON_AIRBNB	13	13	Yes
SUPERHOST_STATUS	No	no	Yes
AIRBNB_RATING_SCORE	5	5	Yes
AIRBNB_NUMBER_OF_REVIEWS	9	9	Yes
TOTAL_PRICE	$4,200	$4,200	Yes
CHECKIN_DATE	12/22/25	12/22/25	Yes
CHECKOUT_DATE	12/27/25	12/27/25	Yes
CANCELLATION_POLICY_SUMMARY	Free cancellation before December 17. Cancel before check-in on December 22 for a partial refund.	Free cancellation before December 17. Cancel before check-in on December 22 for a partial refund.	Yes
Accuracy			93%

The comparison for Test Doc 2 shows high model accuracy, with every field matching except NUMBER_OF_BEDS, where the model undercounted due to inconsistent placement of bed information in the PDF. This demonstrates the model's strong performance and the impact of layout variations on extraction precision.

Test Doc 3

Field	Test Doc 3		
	Actual (from PDF)	Predicted (from model)	Match
NUMBER_OF_+E:PBEDS	The Monarch by Fieldtrip | New Lux Estate w Pool	The Monarch by Fieldtrip — New Lux Estate w Pool	Yes
LOCATION	Yucca Valley, California, United States	Yucca Valley, California, United States	Yes
NUMBER_OF_GUESTS_CAN_ACCOMODATE	10	10	Yes
NUMBER_OF_BEDROOMS	4	4	Yes
NUMBER_OF_BEDS	6	6	Yes
NUMBER_OF_BATHROOMS	3	3	Yes
HOST_NAME	Fieldtrip	Fieldtrip	Yes
HOST_YEARS_ON_AIRBNB	10	10	Yes
SUPERHOST_STATUS	Yes	yes	Yes
AIRBNB_RATING_SCORE	4.86	4.86	Yes
AIRBNB_NUMBER_OF_REVIEWS	72	72	Yes
TOTAL_PRICE	$5,129	$5,129	Yes
CHECKIN_DATE	12/22/25	12/22/25	Yes
CHECKOUT_DATE	12/27/25	12/27/25	Yes
CANCELLATION_POLICY_SUMMARY	Free cancellation for 24 hours. Cancel before December 15 for a partial refund.	Free cancellation for 24 hours. Cancel before December 15 for a partial refund.	Yes
Accuracy			100%

The Test Doc 3 results illustrate the model's strong generalization ability, accurately extracting property details, host information, ratings, pricing, and cancellation policy without errors across all defined fields.

Test Doc 4

Field	Test Doc 4		
	Actual (from PDF)	Predicted (from model)	Match
NUMBER_OF_+E:PBEDS	luxury chateau – free heated pool, wine/hike/beach	luxury chateau – free heated pool, wine/hike/beach	Yes
LOCATION	Fallbrook, California, United States	Fallbrook, California, United States	Yes
NUMBER_OF_GUESTS_CAN_ACCOMODATE	15	15	Yes
NUMBER_OF_BEDROOMS	4	4	Yes
NUMBER_OF_BEDS	7	7	Yes
NUMBER_OF_BATHROOMS	4	4	Yes
HOST_NAME	Aaron	Aaron	Yes
HOST_YEARS_ON_AIRBNB	8	8	Yes
SUPERHOST_STATUS	Yes	yes	Yes
AIRBNB_RATING_SCORE	4.97	4.97	Yes
AIRBNB_NUMBER_OF_REVIEWS	148	148	Yes
TOTAL_PRICE	$5,843	$5,843	Yes
CHECKIN_DATE	12/22/25	12/22/25	Yes
CHECKOUT_DATE	12/27/25	12/27/25	Yes
CANCELLATION_POLICY_SUMMARY	Free cancellation for 24 hours. Cancel before December 15 for a partial refund.	Free cancellation for 24 hours. Cancel before December 15 for a partial refund.	Yes
Accuracy			100%

The comparison results for Test Doc 4 demonstrate the model's strong consistency, accurately interpreting all structured and semi - structured fields despite layout complexity. All predicted values align exactly with the actual PDF data, yielding a flawless extraction.

Test Doc 5

Field	Test Doc 5		
	Actual (from PDF)	Predicted (from model)	Match
NUMBER_OF_+E:PBEDS	NEW! Remodeled Modern Creekside, 30min Yosemite	NEW! Remodeled Modern Creekside, 30min Yosemite	Yes
LOCATION	Oakhurst, California, United States	Oakhurst, California, United States	Yes
NUMBER_OF_GUESTS_CAN_ACCOMODATE	12	12	Yes
NUMBER_OF_BEDROOMS	4	4	Yes
NUMBER_OF_BEDS	9	9	Yes
NUMBER_OF_BATHROOMS	3	3	Yes
HOST_NAME	Benjamin	Benjamin	Yes
HOST_YEARS_ON_AIRBNB	3	3	Yes
SUPERHOST_STATUS	Yes	yes	Yes
AIRBNB_RATING_SCORE	4.93	4.93	Yes
AIRBNB_NUMBER_OF_REVIEWS	46	46	Yes
TOTAL_PRICE	$4,865	$4,865	Yes
CHECKIN_DATE	12/22/25	12/22/25	Yes
CHECKOUT_DATE	12/27/25	12/27/25	Yes
CANCELLATION_POLICY_SUMMARY	Free cancellation for 24 hours. Cancel before December 15 for a partial refund.	Free cancellation for 24 hours. Cancel before December 15 for a partial refund.	Yes
Accuracy			100%

The Test Doc 5 results demonstrate excellent model reliability, where all predicted values align perfectly with the PDF data despite layout complexity, indicating strong generalization across diverse Airbnb listing formats.

Overall, the Document AI model performed exceptionally well across all five test Airbnb documents, achieving near - perfect extraction accuracy. Four out of five test files reached 100% accuracy, and one file achieved 93% accuracy, with the only error occurring in Test Doc 2 where the model predicted the incorrect number of beds due to a layout difference. All other fields - including property details, host metadata, rating and review counts, dates, and pricing - were extracted correctly across the dataset.

Prompt engineering played a significant role in improving the model's performance. Early iterations showed confusion between property ratings and host ratings, as well as issues with strikethrough pricing. After refining entity definitions and updating extraction prompts, the model consistently distinguished between similar elements and reliably selected the correct values. Minor variations in listing titles and cancellation policies were attributed to formatting differences across PDFs rather than extraction errors.

This project demonstrates how Snowflake's Document AI can be used to build a reliable, high - accuracy document processing pipeline for structured data extraction from real - world semi - structured documents. Through prompt optimization, careful field definition, and validation against unseen test files, the model proved robust and capable of generalizing across multiple Airbnb layout variations.